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QUATERNIONIC-KÄHLER MANIFOLDS WITH NONNEGATIVE SECTIONAL CURVATURE

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Abstract. We show that a compact quaternionic-Kähler manifold with positive scalar curvature and nonnegative sectional curvature is isometric to a symmetric space. This extends a classical theorem of Berger.

1. Introduction

Throughout this paper, we assume that (M, g) is a compact quaternionic-Kähler manifold of dimension $4m \geq 8$ with positive scalar curvature. In particular, there exists a parallel subbundle $\mathscr{G} \subset \text{End}(TM)$ of rank 3 with the following property. For each point $p \in M$, we can find linear transformations $I, J, K \in \mathscr{G}_p$ such that $I^2 = J^2 = K^2 = IJK = -\text{id}$ and

$$g(X, Y) = g(IX, IY) = g(JX, JY) = g(KX, KY)$$

for all vectors $X, Y \in T_pM$. We may write

$$\mathscr{G}_p = \{aI + bJ + cK : a, b, c \in \mathbb{R}\}.$$

We define

$$\mathscr{J}_p = \{aI + bJ + cK : a, b, c \in \mathbb{R} \text{ and } a^2 + b^2 + c^2 = 1\} \subset \mathscr{G}_p.$$

By work of Alekseevskii [1] (see also [10], Theorem 3.1), the Riemann curvature tensor of (M, g) can be written in the form $R = R_1 + \kappa R_0$. Here, R_0 is defined by

$$\begin{aligned}
&4\,R_0(X, Y, Z, W) \\
&= g(X, Z)\,g(Y, W) - g(X, W)\,g(Y, Z) \\
&+ 2\,g(IX, Y)\,g(IZ, W) + g(IX, Z)\,g(IY, W) - g(IX, W)\,g(IY, Z) \\
&+ 2\,g(JX, Y)\,g(JZ, W) + g(JX, Z)\,g(JY, W) - g(JX, W)\,g(JY, Z) \\
&+ 2\,g(KX, Y)\,g(KZ, W) + g(KX, Z)\,g(KY, W) - g(KX, W)\,g(KY, Z)
\end{aligned}$$

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for all vectors $X, Y, Z, W \in T_pM$ (cf. [4], equation 14.44). Moreover, R_1 is an algebraic curvature tensor satisfying

$$\begin{aligned} R_1(X, Y, Z, W) &= R_1(X, Y, IZ, IW) \\ &= R_1(X, Y, JZ, JW) \\ &= R_1(X, Y, KZ, KW) \end{aligned}$$

for all vectors $X, Y, Z, W \in T_pM$. To summarize, R_1 satisfies all the algebraic properties of a hyper-Kähler curvature tensor, including the first Bianchi identity. In particular, the Ricci tensor of R_1 vanishes identically. Consequently, the Ricci tensor of (M, g) is given by $(m+2)\kappa\, g$.

By a theorem of Salamon, a compact quaternionic-Kähler manifold with positive scalar curvature is simply connected (see [10], Theorem 6.6). Let $\mathrm{Hol}(M, g)$ denote the holonomy group of (M, g). Since M is simply connected, $\mathrm{Hol}(M, g)$ is a compact, connected Lie subgroup of $\mathrm{SO}(4m)$ (see [4], Corollary 10.47 and Theorem 10.51). Moreover, $\mathrm{Hol}(M, g) \subset \mathrm{Sp}(m) \cdot \mathrm{Sp}(1)$ since (M, g) is a quaternionic-Kähler manifold.

Theorem 1. *Let (M, g) be a compact quaternionic-Kähler manifold of dimension $4m \geq 8$ with positive scalar curvature. We further assume that $R(X, Y, X, Y) + R(X, JY, X, JY) \geq 0$ for all points $p \in M$, all $I, J, K \in \mathscr{J}_p$ satisfying $I^2 = J^2 = K^2 = IJK = -\mathrm{id}$, and all unit vectors $X, Y \in T_pM$ satisfying $g(X, Y) = g(IX, Y) = g(JX, Y) = g(KX, Y) = 0$. If $\mathrm{Hol}(M, g) = \mathrm{Sp}(m) \cdot \mathrm{Sp}(1)$, then $R_1 = 0$ at each point in M.*

Combining Theorem 1 with Berger's holonomy classification theorem, we can draw the following conclusion.

Corollary 2. *Let (M, g) be a compact quaternionic-Kähler manifold of dimension $4m \geq 8$ with positive scalar curvature. We further assume that $R(X, Y, X, Y) + R(X, JY, X, JY) \geq 0$ for all points $p \in M$, all $I, J, K \in \mathscr{J}_p$ satisfying $I^2 = J^2 = K^2 = IJK = -\mathrm{id}$, and all unit vectors $X, Y \in T_pM$ satisfying $g(X, Y) = g(IX, Y) = g(JX, Y) = g(KX, Y) = 0$. Then (M, g) is isometric to a symmetric space.*

Corollary 2 extends earlier work of Berger [3] and Chow and Yang [7].

LeBrun and Salamon [8] have conjectured that a compact quaternionic-Kähler manifold with positive scalar curvature is isometric to a symmetric space. The LeBrun-Salamon conjecture is known to be true in dimension 8 by work of Poon and Salamon [9], and in dimension 12 and 16 by work of Buczyński and Wiśniewski [6].

2. Proof of Theorem 1

In this section, we give the proof of Theorem 1. Throughout this section, we assume that (M, g) is a Riemannian manifold satisfying the assumptions of Theorem 1.

Lemma 3. *We have* $2\,R_1(X,JX,Y,JY) \geq -\kappa$ *for all points* $p \in M$*, all* $I,J,K \in \mathscr{J}_p$ *satisfying* $I^2 = J^2 = K^2 = IJK = -\mathrm{id}$*, and all unit vectors* $X,Y \in T_pM$ *satisfying* $g(X,Y) = g(IX,Y) = g(JX,Y) = g(KX,Y) = 0$*.*

Proof. Fix a point $p \in M$. Suppose that $I,J,K \in \mathscr{J}_p$ satisfy $I^2 = J^2 = K^2 = IJK = -\mathrm{id}$, and let $X,Y \in T_pM$ be unit vectors satisfying $g(X,Y) = g(IX,Y) = g(JX,Y) = g(KX,Y) = 0$. By assumption, $R(X,Y,X,Y) + R(X,JY,X,JY) \geq 0$. Since $R_0(X,Y,X,Y) = R_0(X,JY,X,JY) = \frac{1}{4}$, it follows that

$$R_1(X,Y,X,Y) + R_1(X,JY,X,JY) \geq -\frac{\kappa}{2}.$$

On the other hand, using the first Bianchi identity, we obtain

$$\begin{aligned} R_1(X,JX,Y,JY) &= -R_1(Y,X,JX,JY) - R_1(JX,Y,X,JY) \\ &= R_1(X,Y,X,Y) + R_1(X,JY,X,JY). \end{aligned}$$

Putting these facts together, the assertion follows. This completes the proof of Lemma 3.

Lemma 4. *Suppose that* $p \in M$*, and suppose that* $I,J,K \in \mathscr{J}_p$ *satisfy* $I^2 = J^2 = K^2 = IJK = -\mathrm{id}$*. Then*

$$\begin{aligned} &R_1(Y,V,W,V) + R_1(Y,IV,W,IV) \\ &+ R_1(Y,JV,W,JV) + R_1(Y,KV,W,KV) = 0 \end{aligned}$$

for all $V,Y,W \in T_pM$*.*

Proof. Since R_1 satisfies the first Bianchi identity, we obtain

$$\begin{aligned} R_1(Y,JW,V,JV) &= -R_1(V,Y,JW,JV) - R_1(JW,V,Y,JV) \\ &= R_1(Y,V,W,V) + R_1(Y,JV,W,JV). \end{aligned}$$

Replacing V by IV gives

$$\begin{aligned} -R_1(Y,JW,IV,KV) &= R_1(IV,Y,JW,KV) + R_1(JW,IV,Y,KV) \\ &= R_1(Y,IV,W,IV) + R_1(Y,KV,W,KV). \end{aligned}$$

Since $R_1(Y,JW,V,JV) = R_1(Y,JW,IV,KV)$, the assertion follows.

Lemma 5. *Suppose that* $p \in M$*, and suppose that* $I,J,K \in \mathscr{J}_p$ *satisfy* $I^2 = J^2 = K^2 = IJK = -\mathrm{id}$*. Then*

$$\begin{aligned} &(D_Z R_1)(Y,V,W,V) + (D_Z R_1)(Y,IV,W,IV) \\ &+ (D_Z R_1)(Y,JV,W,JV) + (D_Z R_1)(Y,KV,W,KV) = 0 \end{aligned}$$

and

$$\begin{aligned} &(D^2_{X,Z} R_1)(Y,V,W,V) + (D^2_{X,Z} R_1)(Y,IV,W,IV) \\ &+ (D^2_{X,Z} R_1)(Y,JV,W,JV) + (D^2_{X,Z} R_1)(Y,KV,W,KV) = 0 \end{aligned}$$

for all $V,X,Y,Z,W \in T_pM$*.*

Proof. This follows from Lemma 4.

We next define

$$\begin{aligned}
&T_V^{(0)}(X,Y,Z,W) \\
&= \sum_{l=1}^{4m} R_0(X,Y,V,e_l)\,R_1(Z,W,V,e_l) \\
&+ \sum_{l=1}^{4m} R_0(X,V,Z,e_l)\,R_1(Y,V,W,e_l) - \sum_{l=1}^{4m} R_0(X,V,W,e_l)\,R_1(Y,V,Z,e_l) \\
&- \sum_{l=1}^{4m} R_0(Y,V,Z,e_l)\,R_1(X,V,W,e_l) + \sum_{l=1}^{4m} R_0(Y,V,W,e_l)\,R_1(X,V,Z,e_l)
\end{aligned}$$

and

$$\begin{aligned}
&T_V^{(1)}(X,Y,Z,W) \\
&= \sum_{l=1}^{4m} R_1(X,Y,V,e_l)\,R_1(Z,W,V,e_l) \\
&+ \sum_{l=1}^{4m} R_1(X,V,Z,e_l)\,R_1(Y,V,W,e_l) - \sum_{l=1}^{4m} R_1(X,V,W,e_l)\,R_1(Y,V,Z,e_l) \\
&- \sum_{l=1}^{4m} R_1(Y,V,Z,e_l)\,R_1(X,V,W,e_l) + \sum_{l=1}^{4m} R_1(Y,V,W,e_l)\,R_1(X,V,Z,e_l)
\end{aligned}$$

for all points $p \in M$ and all $V, X, Y, Z, W \in T_pM$. For abbreviation, we put

$$T_V(X,Y,Z,W) = T_V^{(1)}(X,Y,Z,W) + \kappa\,T_V^{(0)}(X,Y,Z,W)$$

for all points $p \in M$ and all $V, X, Y, Z, W \in T_pM$.

We further define

$$\begin{aligned}
&S_V^{(0)}(X,Y,Z,W) \\
&= \sum_{l=1}^{4m} R_0(X,V,e_l,V)\,R_1(e_l,Y,Z,W) + \sum_{l=1}^{4m} R_0(Y,V,e_l,V)\,R_1(X,e_l,Z,W)
\end{aligned}$$

and

$$\begin{aligned}
&S_V^{(1)}(X,Y,Z,W) \\
&= \sum_{l=1}^{4m} R_1(X,V,e_l,V)\,R_1(e_l,Y,Z,W) + \sum_{l=1}^{4m} R_1(Y,V,e_l,V)\,R_1(X,e_l,Z,W)
\end{aligned}$$

for all points $p \in M$ and all $V, X, Y, Z, W \in T_pM$. For abbreviation, we put

$$S_V(X,Y,Z,W) = S_V^{(1)}(X,Y,Z,W) + \kappa\,S_V^{(0)}(X,Y,Z,W)$$

for all points $p \in M$ and all $V, X, Y, Z, W \in T_pM$.

Lemma 6. *We have*

$$\begin{aligned}&(D^2_{X,Z}R_1)(Y,V,W,V) - (D^2_{X,W}R_1)(Y,V,Z,V)\\&- (D^2_{Y,Z}R_1)(X,V,W,V) + (D^2_{Y,W}R_1)(X,V,Z,V)\\&= (D^2_{V,V}R_1)(X,Y,Z,W) + T_V(X,Y,Z,W) - S_V(X,Y,Z,W)\end{aligned}$$

for all $V,X,Y,Z,W \in T_pM$.

Proof. We compute

$$\begin{aligned}&(D^2_{X,V}R_1)(V,Y,Z,W) - (D^2_{V,X}R_1)(Y,V,W,Z)\\&= \sum_{l=1}^{4m} R(X,V,V,e_l)\,R_1(e_l,Y,Z,W) + \sum_{l=1}^{4m} R(X,V,Y,e_l)\,R_1(V,e_l,Z,W)\\&+ \sum_{l=1}^{4m} R(X,V,Z,e_l)\,R_1(V,Y,e_l,W) + \sum_{l=1}^{4m} R(X,V,W,e_l)\,R_1(V,Y,Z,e_l)\end{aligned}$$

and

$$\begin{aligned}&(D^2_{Y,V}R_1)(X,V,Z,W) - (D^2_{V,Y}R_1)(X,V,Z,W)\\&= \sum_{l=1}^{4m} R(Y,V,X,e_l)\,R_1(e_l,V,Z,W) + \sum_{l=1}^{4m} R(Y,V,V,e_l)\,R_1(X,e_l,Z,W)\\&+ \sum_{l=1}^{4m} R(Y,V,Z,e_l)\,R_1(X,V,e_l,W) + \sum_{l=1}^{4m} R(Y,V,W,e_l)\,R_1(X,V,Z,e_l).\end{aligned}$$

Moreover, $R(X,V,Y,e_l) - R(Y,V,X,e_l) = R(X,Y,V,e_l)$ by the first Bianchi identity. Putting these facts together, we obtain

$$\begin{aligned}&(D^2_{X,V}R_1)(V,Y,Z,W) - (D^2_{V,X}R_1)(Y,V,W,Z)\\&+ (D^2_{Y,V}R_1)(X,V,Z,W) - (D^2_{V,Y}R_1)(X,V,Z,W)\\&= \sum_{l=1}^{4m} R(X,Y,V,e_l)\,R_1(V,e_l,Z,W)\\&+ \sum_{l=1}^{4m} R(X,V,Z,e_l)\,R_1(Y,V,W,e_l) - \sum_{l=1}^{4m} R(X,V,W,e_l)\,R_1(Y,V,Z,e_l)\\&- \sum_{l=1}^{4m} R(Y,V,Z,e_l)\,R_1(X,V,W,e_l) + \sum_{l=1}^{4m} R(Y,V,W,e_l)\,R_1(X,V,Z,e_l)\\&- \sum_{l=1}^{4m} R(X,V,e_l,V)\,R_1(e_l,Y,Z,W) - \sum_{l=1}^{4m} R(Y,V,e_l,V)\,R_1(X,e_l,Z,W)\\&= T_V(X,Y,Z,W) - S_V(X,Y,Z,W).\end{aligned}$$

On the other hand, since R_0 is parallel, the covariant derivatives of R_1 satisfy the second Bianchi identity. This implies

$$(D^2_{X,Z}R_1)(Y,V,W,V) - (D^2_{X,W}R_1)(Y,V,Z,V) = (D^2_{X,V}R_1)(Y,V,W,Z),$$

$$-(D^2_{Y,Z}R_1)(X,V,W,V) + (D^2_{Y,W}R_1)(X,V,Z,V) = (D^2_{Y,V}R_1)(X,V,Z,W),$$

and

$$(D^2_{V,V}R_1)(X,Y,Z,W) = (D^2_{V,X}R_1)(V,Y,Z,W) + (D^2_{V,Y}R_1)(X,V,Z,W).$$

From this, the assertion follows. This completes the proof of Lemma 6.

Lemma 7. *Suppose that $p \in M$, and suppose that $I, J, K \in \mathscr{J}_p$ satisfy $I^2 = J^2 = K^2 = IJK = -\text{id}$. Then*

$$\begin{aligned}
0 &= (D^2_{V,V}R_1)(X,Y,Z,W) + (D^2_{IV,IV}R_1)(X,Y,Z,W) \\
&\quad + (D^2_{JV,JV}R_1)(X,Y,Z,W) + (D^2_{KV,KV}R_1)(X,Y,Z,W) \\
&\quad + T_V(X,Y,Z,W) + T_{IV}(X,Y,Z,W) \\
&\quad + T_{JV}(X,Y,Z,W) + T_{KV}(X,Y,Z,W) \\
&\quad - S_V(X,Y,Z,W) - S_{IV}(X,Y,Z,W) \\
&\quad - S_{JV}(X,Y,Z,W) - S_{KV}(X,Y,Z,W)
\end{aligned}$$

for all $V, X, Y, Z, W \in T_pM$ and all $J \in \mathscr{J}_p$.

Proof. This follows by combining Lemma 5 and Lemma 6.

We define

$$\mu = \sup\{R_1(X,JX,X,JX) : p \in M,\, J \in \mathscr{J}_p,\, X \in T_pM,\, |X| = 1\}.$$

The following result follows from work of Berger.

Proposition 8 (cf. M. Berger [3])**.** *We have $\mu = 0$ or $\mu = \kappa$.*

Proof. Let us fix a point $p \in M$, an almost complex structure $J \in \mathscr{J}_p$, and a unit vector $X \in T_pM$ such that $R_1(X,JX,X,JX) = \mu$. Moreover, we can find almost complex structures $I, K \in \mathscr{J}_p$ so that $I^2 = J^2 = K^2 = IJK = -\text{id}$. Let $w_1 = X$ and $w_2 = IX$. Suppose that $w_3, \ldots, w_{2m}$ are vectors in T_pM with the property that $\{w_1, Jw_1, w_2, Jw_2, \ldots, w_{2m}, Jw_{2m}\}$ is an orthonormal basis of T_pM. In view of the maximality of $R_1(X,JX,X,JX)$, we obtain

$$\sum_{\alpha=3}^{2m} (D^2_{w_\alpha,w_\alpha}R_1)(X,JX,X,JX) + \sum_{\alpha=3}^{2m} (D^2_{Jw_\alpha,Jw_\alpha}R_1)(X,JX,X,JX) \leq 0.$$

On the other hand, Lemma 7 implies

$$\begin{aligned}0 &= \sum_{\alpha=3}^{2m}(D^2_{w_\alpha,w_\alpha}R_1)(X,JX,X,JX) + \sum_{\alpha=3}^{2m}(D^2_{Jw_\alpha,Jw_\alpha}R_1)(X,JX,X,JX)\\ &+ \sum_{\alpha=3}^{2m}T_{w_\alpha}(X,JX,X,JX) + \sum_{\alpha=3}^{2m}T_{Jw_\alpha}(X,JX,X,JX)\\ &- \sum_{\alpha=3}^{2m}S_{w_\alpha}(X,JX,X,JX) - \sum_{\alpha=3}^{2m}S_{Jw_\alpha}(X,JX,X,JX).\end{aligned}$$

Putting these facts together, we obtain

$$\begin{aligned}&\sum_{\alpha=3}^{2m}S_{w_\alpha}(X,JX,X,JX) + \sum_{\alpha=3}^{2m}S_{Jw_\alpha}(X,JX,X,JX)\\ &\leq \sum_{\alpha=3}^{2m}T_{w_\alpha}(X,JX,X,JX) + \sum_{\alpha=3}^{2m}T_{Jw_\alpha}(X,JX,X,JX).\end{aligned} \tag{1}$$

Both sides of the inequality (1) are independent of the choice of $w_3,\dots,w_{2m}$. In the following, we choose $w_3,\dots,w_{2m}$ so that

$$R_1(X,JX,w_\alpha,w_\beta) = R_1(X,JX,w_\alpha,Jw_\beta) = 0$$

for all $\alpha,\beta\in\{3,\dots,2m\}$ with $\alpha\neq\beta$. Applying Lemma 9.16 in [5] to the algebraic curvature tensor R_1, we obtain

$$R_1(X,JX,X,w_\beta) = R_1(X,JX,X,Jw_\beta) = 0$$

for all $\beta\in\{2,\dots,2m\}$. Moreover,

$$R_1(X,JX,X,Iw_\beta) = R_1(X,JX,X,JIw_\beta) = 0$$

for all $\beta\in\{3,\dots,2m\}$. Putting these facts together, we conclude that

$$R_1(X,JX,w_\alpha,w_\beta) = R_1(X,JX,w_\alpha,Jw_\beta) = 0$$

for all $\alpha,\beta\in\{1,\dots,2m\}$ with $\alpha\neq\beta$.

We compute

$$\begin{aligned}&T^{(1)}_{w_\alpha}(X,JX,X,JX) + T^{(1)}_{Jw_\alpha}(X,JX,X,JX)\\ &= 2\sum_{\beta=1}^{2m}R_1(X,JX,w_\alpha,w_\beta)^2 + 2\sum_{\beta=1}^{2m}R_1(X,JX,w_\alpha,Jw_\beta)^2\\ &- 8\sum_{\beta=1}^{2m}R_1(X,w_\alpha,JX,w_\beta)\,R_1(JX,w_\alpha,X,w_\beta)\\ &- 8\sum_{\beta=1}^{2m}R_1(X,w_\alpha,JX,Jw_\beta)\,R_1(JX,w_\alpha,X,Jw_\beta)\end{aligned}$$

for each $\alpha \in \{3, \ldots, 2m\}$. Using the inequality

$$\begin{aligned}
&- 8 \sum_{\beta=1}^{2m} R_1(X, w_\alpha, JX, w_\beta)\, R_1(JX, w_\alpha, X, w_\beta) \\
&\leq 2 \sum_{\beta=1}^{2m} (R_1(X, w_\alpha, JX, w_\beta) - R_1(JX, w_\alpha, X, w_\beta))^2 \\
&= 2 \sum_{\beta=1}^{2m} R_1(X, JX, w_\alpha, w_\beta)^2
\end{aligned}$$

and the inequality

$$\begin{aligned}
&- 8 \sum_{\beta=1}^{2m} R_1(X, w_\alpha, JX, Jw_\beta)\, R_1(JX, w_\alpha, X, Jw_\beta) \\
&\leq 2 \sum_{\beta=1}^{2m} (R_1(X, w_\alpha, JX, Jw_\beta) - R_1(JX, w_\alpha, X, Jw_\beta))^2 \\
&= 2 \sum_{\beta=1}^{2m} R_1(X, JX, w_\alpha, Jw_\beta)^2,
\end{aligned}$$

we deduce that

$$\begin{aligned}
&T^{(1)}_{w_\alpha}(X, JX, X, JX) + T^{(1)}_{Jw_\alpha}(X, JX, X, JX) \\
&\leq 4 \sum_{\beta=1}^{2m} R_1(X, JX, w_\alpha, w_\beta)^2 + 4 \sum_{\beta=1}^{2m} R_1(X, JX, w_\alpha, Jw_\beta)^2
\end{aligned} \tag{2}$$

for each $\alpha \in \{3, \ldots, 2m\}$. We sum both sides of the inequality (2) over $\alpha \in \{3, \ldots, 2m\}$. We recall that $R_1(X, JX, w_\alpha, w_\beta) = R_1(X, JX, w_\alpha, Jw_\beta) = 0$ for all $\alpha, \beta \in \{1, \ldots, 2m\}$ with $\alpha \neq \beta$. This gives

$$\begin{aligned}
&\sum_{\alpha=3}^{2m} T^{(1)}_{w_\alpha}(X, JX, X, JX) + \sum_{\alpha=3}^{2m} T^{(1)}_{Jw_\alpha}(X, JX, X, JX) \\
&\leq 4 \sum_{\alpha=3}^{2m} R_1(X, JX, w_\alpha, Jw_\alpha)^2.
\end{aligned} \tag{3}$$

Moreover, using Lemma 4, we compute

$$\sum_{\alpha=3}^{2m} S^{(1)}_{w_\alpha}(X, JX, X, JX) + \sum_{\alpha=3}^{2m} S^{(1)}_{Jw_\alpha}(X, JX, X, JX) = 0. \tag{4}$$

In the next step, we recall a result from [5]. It follows from Proposition 9.23 in [5] that $B(R_1, R_0) = 0$, where $B(R_1, R_0)$ denotes the algebraic expression

defined on p. 141 of [5]. This implies

$$\sum_{\alpha=1}^{2m} T^{(0)}_{w_\alpha}(X,JX,X,JX) + \sum_{\alpha=1}^{2m} T^{(0)}_{Jw_\alpha}(X,JX,X,JX) = 0. \tag{5}$$

Using the formula for R_0, we compute

$$T^{(0)}_X(X,JX,X,JX) = R_1(X,JX,X,JX),$$
$$T^{(0)}_{JX}(X,JX,X,JX) = R_1(JX,X,JX,X),$$
$$T^{(0)}_{IX}(X,JX,X,JX) = R_1(X,IX,X,IX) + R_1(JX,IX,JX,IX),$$
$$T^{(0)}_{KX}(X,JX,X,JX) = R_1(X,KX,X,KX) + R_1(JX,KX,JX,KX).$$

We add these four identities. Using Lemma 4, we deduce that

$$\sum_{\alpha=1}^{2} T^{(0)}_{w_\alpha}(X,JX,X,JX) + \sum_{\alpha=1}^{2} T^{(0)}_{Jw_\alpha}(X,JX,X,JX) = 0. \tag{6}$$

Subtracting (6) from (5) gives

$$\sum_{\alpha=3}^{2m} T^{(0)}_{w_\alpha}(X,JX,X,JX) + \sum_{\alpha=3}^{2m} T^{(0)}_{Jw_\alpha}(X,JX,X,JX) = 0. \tag{7}$$

Since the Ricci tensor of R_0 is given by $(m+2)\,g$, we obtain

$$\begin{aligned} &\sum_{\alpha=1}^{2m} S^{(0)}_{w_\alpha}(X,JX,X,JX) + \sum_{\alpha=1}^{2m} S^{(0)}_{Jw_\alpha}(X,JX,X,JX) \\ &= (2m+4)\,R_1(X,JX,X,JX) = (2m+4)\mu. \end{aligned} \tag{8}$$

Moreover,

$$\begin{aligned} &\sum_{\alpha=1}^{2} S^{(0)}_{w_\alpha}(X,JX,X,JX) + \sum_{\alpha=1}^{2} S^{(0)}_{Jw_\alpha}(X,JX,X,JX) \\ &= 6\,R_1(X,JX,X,JX) = 6\mu. \end{aligned} \tag{9}$$

Subtracting (9) from (8) gives

$$\sum_{\alpha=3}^{2m} S^{(0)}_{w_\alpha}(X,JX,X,JX) + \sum_{\alpha=3}^{2m} S^{(0)}_{Jw_\alpha}(X,JX,X,JX) = (2m-2)\mu. \tag{10}$$

Combining (1), (3), (4), (7), and (10), we conclude that

$$(2m-2)\kappa\mu \le 4\sum_{\alpha=3}^{2m} R_1(X,JX,w_\alpha,Jw_\alpha)^2. \tag{11}$$

On the other hand, it follows from Lemma 3 that

$$2\,R_1(X,JX,w_\alpha,Jw_\alpha) \ge -\kappa$$

and

$$2\,R_1(X,JX,Iw_\alpha,JIw_\alpha) \ge -\kappa$$

for each $\alpha \in \{3, \ldots, 2m\}$. Therefore,

$$2\,|R_1(X, JX, w_\alpha, Jw_\alpha)| \leq \kappa \tag{12}$$

for each $\alpha \in \{3, \ldots, 2m\}$. Applying Lemma 9.16 in [5] to the algebraic curvature tensor R_1, we obtain

$$2\,R_1(X, JX, w_\alpha, Jw_\alpha) \leq R_1(X, JX, X, JX) = \mu$$

and

$$2\,R_1(X, JX, Iw_\alpha, JIw_\alpha) \leq R_1(X, JX, X, JX) = \mu$$

for each $\alpha \in \{3, \ldots, 2m\}$. Consequently,

$$2\,|R_1(X, JX, w_\alpha, Jw_\alpha)| \leq \mu \tag{13}$$

for each $\alpha \in \{3, \ldots, 2m\}$. Combining (11) and (12), we obtain $(2m-2)\kappa\mu \leq (2m-2)\kappa^2$. Combining (11) and (13), we obtain $(2m-2)\kappa\mu \leq (2m-2)\mu^2$. Since $m \geq 2$, $\kappa > 0$, and $\mu \geq 0$, it follows that either $\mu = 0$ or $\mu = \kappa$. This completes the proof of Proposition 8.

Remark 9. Suppose that $R(X, Y, X, Y) + R(X, JY, X, JY) > 0$ for all points $p \in M$, all $I, J, K \in \mathscr{J}_p$ satisfying $I^2 = J^2 = K^2 = IJK = -\text{id}$, and all unit vectors $X, Y \in T_pM$ satisfying $g(X, Y) = g(IX, Y) = g(JX, Y) = g(KX, Y) = 0$. In this case, the strict inequality holds in (12). The inequality (11) then implies $(2m-2)\kappa\mu < (2m-2)\kappa^2$. On the other hand, combining (11) and (13) gives $(2m-2)\kappa\mu \leq (2m-2)\mu^2$. Since $m \geq 2$, $\kappa > 0$, and $\mu \geq 0$, it follows that $\mu = 0$ and R_1 vanishes identically. This recovers Berger's classical theorem [3].

We now continue with the proof of Theorem 1. We consider the bundle $\text{End}(TM) \oplus TM$. Let $\pi : \text{End}(TM) \oplus TM \to M$ denote the associated bundle projection. We shall define a subset $\mathscr{E}$ of the total space of this bundle. The set $\mathscr{E}$ consists of all triplets (p, J, X) with the property that $p \in M$, $J \in \mathscr{J}_p$, $X \in T_pM$, and $|X| = 1$. Clearly, $\mathscr{E}$ is a compact submanifold of dimension $8m+1$. Moreover, the set $\mathscr{E}$ is invariant under parallel transport.

We consider the function

$$\varphi : \mathscr{E} \to \mathbb{R}, \quad (p, J, X) \mapsto \kappa - R_1(X, JX, X, JX).$$

Lemma 10. *Let U be an open subset of M which is diffeomorphic to B^{4m}. Let $\{e_1, \ldots, e_{4m}\}$ be a local orthonormal frame defined on U. For each $l \in \{1, \ldots, 4m\}$, we lift the vector field e_l on U to a horizontal vector field on $\mathscr{E} \cap \pi^{-1}(U)$, which we denote by $\mathscr{X}_l$. Moreover, for each $l \in \{1, \ldots, 4m\}$, we lift the vector field $D_{e_l}e_l$ on U to a horizontal vector field on $\mathscr{E} \cap \pi^{-1}(U)$, which we denote by $\mathscr{Y}_l$. Then*

$$\begin{aligned}&\sum_{l=1}^{4m} \mathscr{X}_l(\mathscr{X}_l(\varphi)) - \sum_{l=1}^{4m} \mathscr{Y}_l(\varphi) \\ &\leq -C \inf_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} (D^2\varphi)(\xi, \xi) + C \sup_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} d\varphi(\xi) + C\,|\varphi|\end{aligned}$$

at each point $(p, J, X) \in \mathscr{E} \cap \pi^{-1}(U)$. *Here,* $\mathscr{V}_{(p,J,X)} \subset T_{(p,J,X)}\mathscr{E}$ *denotes the vertical space at the point* $(p, J, X) \in \mathscr{E}$. *Moreover,* C *is a large positive constant.*

Proof. The Ricci tensor of (M, g) is given by $(m+2)\kappa\, g$. Using Proposition 2.11 in [5], we obtain

$$\Delta R + Q(R) = (2m+4)\kappa\, R.$$

Here,

$$\Delta R = \sum_{l=1}^{4m} D^2_{e_l,e_l} R = \sum_{l=1}^{4m} D_{e_l} D_{e_l} R - \sum_{l=1}^{4m} D_{D_{e_l} e_l} R$$

denotes the Laplacian of the Riemann curvature tensor, and $Q(R)$ denotes the algebraic expression defined on p. 53 of [5]. Clearly, $\Delta R = \Delta R_1$ since R_0 is parallel. Moreover, Proposition 9.23 in [5] implies that $Q(R) = Q(R_1) + \kappa^2\, Q(R_0) = Q(R_1) + (2m+4)\kappa^2\, R_0$. Putting these facts together, we conclude that

$$\Delta R_1 + Q(R_1) = (2m+4)\kappa\, R_1. \tag{14}$$

For each $l \in \{1, \ldots, 4m\}$, we have

$$\mathscr{X}_l(\mathscr{X}_l(\varphi)) = -(D_{e_l} D_{e_l} R_1)(X, JX, X, JX)$$

and

$$\mathscr{Y}_l(\varphi) = -(D_{D_{e_l} e_l} R_1)(X, JX, X, JX)$$

at each point $(p, J, X) \in \mathscr{E} \cap \pi^{-1}(U)$. This implies

$$\sum_{l=1}^{4m} \mathscr{X}_l(\mathscr{X}_l(\varphi)) - \sum_{l=1}^{4m} \mathscr{Y}_l(\varphi) = -\Delta R_1(X, JX, X, JX) \tag{15}$$

at each point $(p, J, X) \in \mathscr{E} \cap \pi^{-1}(U)$. Combining (14) and (15), we obtain

$$\begin{aligned} &\sum_{l=1}^{4m} \mathscr{X}_l(\mathscr{X}_l(\varphi)) - \sum_{l=1}^{4m} \mathscr{Y}_l(\varphi) \\ &= Q(R_1)(X, JX, X, JX) - (2m+4)\kappa\, R_1(X, JX, X, JX) \end{aligned} \tag{16}$$

at each point $(p, J, X) \in \mathscr{E} \cap \pi^{-1}(U)$.

In order to estimate the term on the right hand side of (16), we adapt the proof of Theorem 9.24 in [5]. Let us fix a triplet $(p, J, X) \in \mathscr{E} \cap \pi^{-1}(U)$. As above, we put $w_1 = X$ and $w_2 = IX$. We can find vectors $w_3, \ldots, w_{2m} \in T_pM$ with the property that $\{w_1, Jw_1, w_2, Jw_2, \ldots, w_{2m}, Jw_{2m}\}$ is an orthonormal basis of T_pM and

$$R_1(X, JX, w_\alpha, w_\beta) = R_1(X, JX, w_\alpha, Jw_\beta) = 0 \tag{17}$$

for all $\alpha, \beta \in \{3, \ldots, 2m\}$ with $\alpha \neq \beta$.

By adapting the proof of Lemma 9.16 in [5], we can show that

$$|R_1(X, JX, X, w_\beta)| \leq C \sup_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} d\varphi(\xi) \tag{18}$$

and

$$|R_1(X, JX, X, Jw_\beta)| \leq C \sup_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} d\varphi(\xi) \tag{19}$$

for all $\beta \in \{2, \ldots, 2m\}$. An analogous argument gives

$$|R_1(X, JX, X, Iw_\beta)| \leq C \sup_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} d\varphi(\xi)$$

and

$$|R_1(X, JX, X, JIw_\beta)| \leq C \sup_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} d\varphi(\xi)$$

for all $\beta \in \{3, \ldots, 2m\}$. These inequalities can be rewritten as

$$|R_1(X, JX, IX, w_\beta)| \leq C \sup_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} d\varphi(\xi) \tag{20}$$

and

$$|R_1(X, JX, IX, Jw_\beta)| \leq C \sup_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} d\varphi(\xi) \tag{21}$$

for all $\beta \in \{3, \ldots, 2m\}$. Combining (17), (18), (19), (20), and (21), we conclude that

$$|R_1(X, JX, w_\alpha, w_\beta)| \leq C \sup_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} d\varphi(\xi) \tag{22}$$

and

$$|R_1(X, JX, w_\alpha, Jw_\beta)| \leq C \sup_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} d\varphi(\xi) \tag{23}$$

for all $\alpha, \beta \in \{1, \ldots, 2m\}$ with $\alpha \neq \beta$. Moreover, by adapting the proof of Lemma 9.16 in [5], we can show that

$$\begin{aligned} 2\, R_1(X, JX, w_\alpha, Jw_\alpha) &\leq R_1(X, JX, X, JX) \\ &\quad - C \inf_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} (D^2\varphi)(\xi, \xi) \\ &\quad + C \sup_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} d\varphi(\xi) \end{aligned} \tag{24}$$

for each $\alpha \in \{3, \ldots, 2m\}$. An analogous argument gives

$$\begin{aligned} 2\, R_1(X, JX, Iw_\alpha, JIw_\alpha) &\leq R_1(X, JX, X, JX) \\ &\quad - C \inf_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} (D^2\varphi)(\xi, \xi) \\ &\quad + C \sup_{\xi \in \mathscr{V}_{(p,J,X)}, |\xi| \leq 1} d\varphi(\xi) \end{aligned}$$

for each $\alpha \in \{3, \ldots, 2m\}$. This inequality can be rewritten as

$$\begin{aligned} -2\, R_1(X, JX, w_\alpha, Jw_\alpha) &\leq R_1(X, JX, X, JX) \\ &\quad - C \inf_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} (D^2\varphi)(\xi, \xi) \\ &\quad + C \sup_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} d\varphi(\xi) \end{aligned} \tag{25}$$

for each $\alpha \in \{3, \ldots, 2m\}$. Combining (24) and (25), we conclude that

$$\begin{aligned} 2\, |R_1(X, JX, w_\alpha, Jw_\alpha)| &\leq R_1(X, JX, X, JX) \\ &\quad - C \inf_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} (D^2\varphi)(\xi, \xi) \\ &\quad + C \sup_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} d\varphi(\xi) \end{aligned} \tag{26}$$

for each $\alpha \in \{3, \ldots, 2m\}$.

Proposition 9.15 in [5] gives

$$\begin{aligned} Q(R_1)(X, JX, X, JX) &\leq -2\, R_1(X, JX, X, JX)^2 \\ &\quad + 4 \sum_{\alpha,\beta=1}^{2m} R_1(X, JX, w_\alpha, w_\beta)^2 \\ &\quad + 4 \sum_{\alpha,\beta=1}^{2m} R_1(X, JX, w_\alpha, Jw_\beta)^2. \end{aligned}$$

Using (22) and (23), we obtain

$$\begin{aligned} Q(R_1)(X, JX, X, JX) &\leq 6\, R_1(X, JX, X, JX)^2 \\ &\quad + 4 \sum_{\alpha=3}^{2m} R_1(X, JX, w_\alpha, Jw_\alpha)^2 \\ &\quad + C \sup_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} d\varphi(\xi). \end{aligned}$$

Using (26), we conclude that

$$\begin{aligned} Q(R_1)(X, JX, X, JX) &\leq (2m+4)\, R_1(X, JX, X, JX)^2 \\ &\quad - C \inf_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} (D^2\varphi)(\xi, \xi) \\ &\quad + C \sup_{\xi \in \mathscr{V}_{(p,J,X)},\, |\xi| \leq 1} d\varphi(\xi). \end{aligned} \tag{27}$$

Finally,

$$(2m+4)\, R_1(X, JX, X, JX)^2 \leq (2m+4)\kappa\, R_1(X, JX, X, JX) + C\, |\varphi| \tag{28}$$

by definition of φ. Combining (16), (27), and (28), the assertion follows.

Proposition 11. *The function $\varphi : \mathscr{E} \to \mathbb{R}$ is nonnegative. Moreover, the set $\mathscr{F} = \{\varphi = 0\} \subset \mathscr{E}$ is invariant under parallel transport.*

Proof. It follows from Proposition 8 that $\mu \leq \kappa$. This implies $\inf_{\mathscr{E}} \varphi = \kappa - \mu \geq 0$. This proves the first statement. To prove the second statement, suppose that U is an open subset of M which is diffeomorphic to B^{4m}. Using Lemma 10 and the strict maximum principle (cf. [5], Corollary 9.7), we conclude that every smooth horizontal curve in $\mathscr{E} \cap \pi^{-1}(U)$ that starts at a point in $\mathscr{F} \cap \pi^{-1}(U)$ must be contained in $\mathscr{F} \cap \pi^{-1}(U)$. From this, we deduce that every smooth horizontal curve in $\mathscr{E}$ that starts at a point in $\mathscr{F}$ must be contained in $\mathscr{F}$. This completes the proof of Proposition 11.

Proposition 12. *We have $\mu = 0$.*

Proof. We argue by contradiction. Suppose that $\mu \neq 0$. Using Proposition 8, we obtain $\mu = \kappa$. Consequently, we can find a point $p \in M$, an almost complex structure $J \in \mathscr{J}_p$, and a unit vector $X \in T_pM$ such that $R_1(X, JX, X, JX) = \kappa$. Moreover, we can find almost complex structures $I, K \in \mathscr{J}_p$ so that $I^2 = J^2 = K^2 = IJK = -\text{id}$.

Let Z be an arbitrary unit vector in T_pM. Since $\text{Sp}(m)$ acts transitively on the unit sphere S^{4m-1}, we can find a linear isometry $L : T_pM \to T_pM$ which commutes with I, J, K and satisfies $LX = Z$. Since $\text{Hol}(M, g) = \text{Sp}(m) \cdot \text{Sp}(1)$, there exists a piecewise smooth path $\gamma : [0, 1] \to M$ such that $\gamma(0) = \gamma(1) = p$ and $P_\gamma = L$, where $P_\gamma : T_{\gamma(0)}M \to T_{\gamma(1)}M$ denotes the parallel transport along γ. By Proposition 11, the set $\mathscr{F}$ is is invariant under parallel transport. Since $(p, J, X) \in \mathscr{F}$, it follows that $(p, LJL^{-1}, LX) \in \mathscr{F}$. Thus, $(p, J, Z) \in \mathscr{F}$. In other words, $R_1(Z, JZ, Z, JZ) = \kappa$.

To summarize, we have shown that $R_1(Z, JZ, Z, JZ) = \kappa$ for every unit vector $Z \in T_pM$. Since $\kappa > 0$, it follows that the scalar curvature of R_1 at the point p is strictly positive. This contradicts the fact that the scalar curvature of R_1 vanishes identically. This completes the proof of Proposition 12.

Using Proposition 12, it is easy to see that $R_1 = 0$ at each point in M. This completes the proof of Theorem 1.

3. Proof of Corollary 2

In this final section, we give the proof of Corollary 2. Suppose that (M, g) is a Riemannian manifold satisfying the assumptions of Corollary 2. Since (M, g) is a quaternionic-Kähler manifold with positive scalar curvature, M is simply connected (see [10], Theorem 6.6). As above, we denote by $\text{Hol}(M, g) \subset \text{Sp}(m) \cdot \text{Sp}(1)$ the holonomy group of (M, g). We distinguish two cases:

Case 1: Suppose that $\text{Hol}(M, g) = \text{Sp}(m) \cdot \text{Sp}(1)$. In this case, Theorem 1 implies that $R_1 = 0$ at each point in M. Thus, the Riemann curvature tensor of (M, g) is parallel, and (M, g) is isometric to a symmetric space.

Case 2: Suppose that $\mathrm{Hol}(M,g) \subsetneq \mathrm{Sp}(m) \cdot \mathrm{Sp}(1)$. In this case, we obtain $\dim \mathrm{Hol}(M,g) < \dim(\mathrm{Sp}(m) \cdot \mathrm{Sp}(1))$. In particular, $\dim \mathrm{Hol}(M,g) < \dim \mathrm{U}(2m)$ and $\dim \mathrm{Hol}(M,g) < \dim \mathrm{SO}(4m)$. Since (M,g) is a quaternionic-Kähler manifold with positive scalar curvature, it follows from a theorem of Berger that (M,g) is irreducible (see [2] or [4], Theorem 14.45(b)). Using Berger's holonomy classification theorem (cf. [4], Corollary 10.92), we conclude that (M,g) is isometric to a symmetric space. This completes the proof of Corollary 2.

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